LAB 3 – POWER BI DESKTOP

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- Power BI Desktop – data processing and modelling in PBI
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TASK 1 – WIDE TABLE APPROACH

Wide Table approach is a design approach, in which all the data from the OLTP relational schema is joined into one wide denormalized table. As a result, this approach is commonly called the "One Big Table" (OBT for short, this is also probably the more popular name of this design approach). In short, this method, characterized by its simplicity, involves storing data in a single, expansive table, offering a much-simplified data model by significantly reducing the number of tables that need to be governed and updated.

Previously used approach allowed us to analyse data in a single table, or multiple tables using some of the relationships defined between the data. Moreover, in all cases we used the data as-is — directly the way the data is defined and stored within the operational database. Now, let us try to analyse data using Pivot Tables, but first we are going to prepare it – transform to a more suitable form. To do so, we will follow a very simple, somewhat naïve, process: identifying needed data, creating dedicated view and preparing a resultant pivot table.

Using a Power Query Editor prepare data in a OBT structure, where all data is in a single wide table. Before implementation, please first read the entire task.

1. First focus on identifying main tables and important attributes related to products. Look at the data dictionary and investigate attributes available in product-related tables (start analysis from Production.Product) – consider using attributes that are useful in our main goal:
    a. Do we need information about all products? Be able to justify your selection (or lack) of certain attributes.
    b. At least, use the following product details: product name, category name, subcategory name, product's catalogue price, product's color, product's weight category.
2. Next focus on identifying main tables and attributes related to sales representatives. Look at the data dictionary and investigate attributes available in order-related tables:
    a. Just focus on very basic information – include first name, last name and full name of the sales representative.
3. Next focus on identifying main tables and attributes related to time/dates. Look at the data dictionary and investigate attributes available in order-related tables:
    a. Do we need information about all dates?
    b. Identify needed attributes representing time/dates – like month, year. Be able to justify your selection (or lack) of attributes.
4. Next focus on identifying main tables and attributes related to orders (from product and time perspective). Look at the data dictionary and investigate attributes available in order-related tables:
    a. Do we need information about all orders?
    b. Identify needed performance measures – like sales amount, sales volume, profit – that can be used to analyse and measure the performance of sales process. Be able to justify your selection (or lack) of attributes.
5. Prepare a single query which provides basic information about product's sales – covering all user needs:
    a. Note that you need to include product id, product name, product category name and product subcategory name, product color, product weight. Consider including other attributes (use results from point 1). Focus on order's date.
    b. Note that you need to include sales representative information. Consider including all required attributes (use results from point 2)
    c. Note that you need to include time/date information. Consider including all required time/date indicators (use results from point 3)

d. Note that you need to include sales amount and order quantity. Consider including other performance measure (use results from point 4)

Let us now look at some reporting capabilities of the prepared data model and the Power BI Desktop tool (PBI for short). We plan to use the order's data to create a simple set of visualisations to aid to analysis of sales from two basic points of view – focusing on product and customer. In particular, you are now the BI user, and your goal is to consume the available data (prepared in the previous task) to gather some insights about the performance of individual sales representatives. Consider a situation, in which you are trying to identify the most prominent sales representatives and their sales focus.

Use the prepared data model in Power BI Desktop and create a data visualisation for each of the following user requirements (extended from task 1 from laboratory assignment 2):

1. On a new page, please prepare a report to show basic order data in the form (exact):
    a. Order('Lastname, Firstname', Year, Month, ProductCategory, TotalSalesValue).
        i. Sale's reps 'Lastname, firstname' (should be a single column) with proper aggregates per year, month, category.
    b. If this is not possible – answer why not.
2. On a new page, please prepare a dashboard (a single report page with multiple arranged and interconnected visualisations) to help identifying **the most prominent sales representatives** – focus on total sales amount, total quantity of sold products, and total discount. Use time dependencies here. If there are some issues with certain elements, please comment.
    a. Please try answering the following requirements - if this is not possible – answer why not:
        i. Top 5 most prominent sales reps each year.
        ii. Top 5 least prominent sales reps each year.
        iii. Top 10 most prominent sales reps as compared to a given year and a previous year.
        iv. Compare individual sales reps performance in the beginning of the year with that in the end of the year.
        v. Compare individual sales reps performance in the beginning of the month with that in the end of the month.
        vi. Sales reps with better performance on before holidays than after holidays.
3. On a new page, please prepare a dashboard (a single report page with multiple arranged and interconnected visualisations) to help identifying **the most prominent sales representatives**. Use product dependencies here – attributes like subcategories, categories, price, or other product's attributes. Think about possible measures and implement the ones you find most valuable.
    a. Please try answering the following requirements - if this is not possible – answer why not:
        i. Top 5 most prominent sales reps each category/subcategory.
        ii. Top 5 least prominent sales reps each category/subcategory.
        iii. Top 10 most prominent sales reps for expensive products (with price more than 800USD).
        iv. Compare individual sales reps performance for expensive products and inexpensive products.
        v. Compare individual sales reps performance for "frame" products (look at productnumber attribute).
        vi. Top sales reps for "dark" colors (you can specify the semantics of dark colors – it cannot be a single color).
4. On a new page, please prepare a dashboard (a single report page with arranged and interconnected collection of visualisations) to help identifying **the most prominent product categories for a specified sales representative** – focus on total sales amount, total quantity of sold products, and total discount. Use time dependencies here. If there are some issues with certain elements, please comment.
    a. Top 1 most prominent category for expensive products (with price more than 800USD).
    b. Compare performance for expensive products and inexpensive products.
    c. Show performance for "frame" products (look at productnumber attribute).
    d. Show performance for "dark" colors (you can specify the semantics of dark colors – it cannot be a single color).

As the result, please use Power BI Desktop file with reports.

TASK 2 – DIMENSIONAL DATA MODEL IN POWER BI DESKTOP

Let us now look at some data transformation capabilities of Power BI Desktop tool (PBI for short). We plan to use the sales order's data, extend it to include information about products, time and sales representatives. In the next task you'll create a simple set of visualisations to aid to analysis of sales from these three basic points of view – focusing on time, product, sales representative.

We are aiming to define a model like the one depicted in Figure 1. The model there focuses on Time and Product analysis, where Sales represents all individual sales orders (on the level of order item), their measurements (like item sales amount, item quantity, item profit, etc.) and links to: individual time points and individual products. Time represents all individual time point – details about individual dates (like month name, year, etc.). Product represents all individual products – details about individual products (like name, price, etc.).

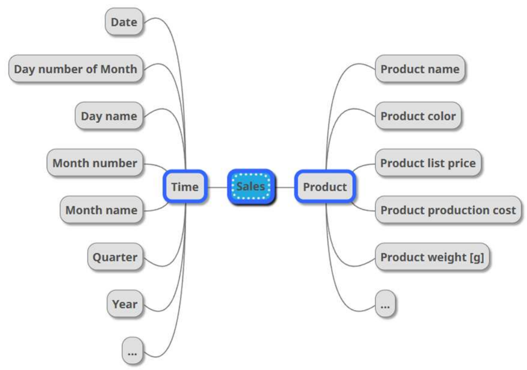

Figure 1. Overview of the resultant data model

Using the Adventure Works database use Power BI Desktop to connect to the data and transform data to the required model. Each element of the model should be represented as a separate query in Power Query Editor – you can obviously use multiple auxiliary queries to get the needed data (later hide these queries from the Report view, to create a clean user experience). Alternatively, you can use a different toolset to prepare the data (series of SQL statements, Python scripts, etc.) – in this approach store the final model in SQL Server (as a new schema/database; remember to create proper primary/foreign keys), connect to this model from PBI, and proceed directly to step 6.

1. Connect to raw data in AdventureWorks database from Power BI Desktop. If needed define proper relationships between PBI queries.
2. Prepare a PBI query that results in a dimension representing products. Please remember that we want to have a unique set of all products. Please include product's name, subcategory name, category name, color, weight, size and list price.
3. Prepare a PBI query that results in a dimension representing sales representatives, do not use dedicated view in AdventureWorks database. In the resultant dataset each row should represent a unique salesperson. Please make sure that first name, last name, full name, address, city, state/province, country name attributes are present.
4. Prepare a PBI query that results in a dimension representing dates. In the resultant dataset each row should represent a unique day. Please make sure that date, day name of week, day of month, month, month name, quarter, year attributes are present.
5. Prepare a PBI query that results in a table representing sales. In the resultant dataset each row should represent a unique sales item (sold order item). Please make sure that basic performance metrics are available – total sales amount, total order quantity, discount value.
6. Define proper relationships between all the newly created PBI queries.

Let us now look at some reporting capabilities of the prepared data model and the Power BI Desktop tool (PBI for short). We plan to use the order's data to create a simple set of visualisations to aid to analysis of sales from two basic points of view – focusing on product and customer. In particular, you are now the BI user, and your goal is to consume the available data (prepared in the previous task) to gather some insights about the performance of individual sales representatives. Consider a situation, in which you are trying to identify the most prominent sales representatives and their sales focus.

Using the prepared data model in Power BI Desktop to generate a set of basic analytical reports (use multiple report pages):

7. Repeat all analytical requirements from task 1.

TASK 3 – COMPARISON AND EXTENDING THE DATA MODEL

Please compare the approaches used in task 2 (laboratory assignment 2), task 1 (laboratory assignment 3 – this one), and task 2 (laboratory assignment 3 – this one). Write a short paragraph about this – it can be a text file, that you'll present during the next class.

(*) Could you explain (in more details) what changes are needed to the process to allow the end user to perform the following analysis: "Which product categories sale best before Christmas?".
*Note that **no implementation** is required here.*

TUTORIAL 1 - DATA PROFILING AND QUALITY ASSESMENT

Let us now look at some data profiling capabilities of the Power BI Desktop tool. The focus is on analysis, determination of characteristics, and evaluation of a part of your sample dataset.

Your task here is to study the dataset, understand and describe the domain at hand, and asses the quality of the data. As such, you are focusing on a form of basic data discovery and data profiling steps. Data discovery describes processes of developing certain level of understanding of the domain and the data within the dataset, typically as the initial step before data integration and/or data analysis. During profiling, you focus on creating profiles that help analysing the content, quality, and structure of data sources. As a part of the profiling process, you discover the metadata of data sources. You use different profiles for different types of data analysis, such as a column value distribution profile, column statistic profile, primary key discovery, foreign key discovery, and data domain discovery. Data profiling enables you to assess the quality of your source data before you use it in data warehousing or other data integration scenarios. In short, data profiling analyses the content, structure, and relationships within data to uncover patterns and rules, inconsistencies, anomalies, and redundancies. Finally, data discovery and data profiling offer businesses a way to make their data clean, easily understandable, and user-friendly. A comprehensive solution should be able to be used by all members of the business.

In the process, you typically use a set of dedicated data profiling tools. For instance, you can perform basic data profiling using Power BI Desktop (for details see https://learn.microsoft.com/en-us/power-query/data-profiling-tools). The process is straightforward, you need to connect to the dataset (Get Data, using a dedicated connector) and open the Power Query Editor (available at the transformation stage, you can use Transform Data menu option). Within the editor you have access to basic data profiling tools in the View > Data Preview part of the menu ribbon. Tools allow you to investigate column quality, column distribution and column profile.

Similarly, you can use Tableau Prep (for details see https://help.tableau.com/current/prep/en-us/prep_explore.htm). The process is very intuitive and simple. In short, you just need to connect to the dataset (for instance using a Text file connection) and add a Clean step in the process flow. In the Clean step you have access to basic visualisations that support the study of size details and access to a series of interactive histograms to study value distributions. Moreover, you can use a more advanced mechanism of Data Roles (ability to check values against predefined lists of curated values). The ability to profile data in Tableau is limited to attribute analysis, which focuses on independent analysis of individual columns. In short, it is hard to investigate relations between attributes, like functional dependencies (between columns).

A solid alternative, to data profiling is the Data Profiling task from SQL Server Integration Services project (available in Visual Studio – we will cover this in details later). This tool provides more complex data profiling functionality, where a data profile is understood as a basic collection of aggregate statistics about data. For a short introduction how to perform basic data profiling for the required data (Use Data Profiling Task in Control Flow) look at: https://docs.microsoft.com/en-us/sql/integration-services/control-flow/setup-of-the-data-profiling-task and https://docs.microsoft.com/en-us/sql/integration-services/control-flow/data-profiling-task. Data profiling task generates a profile file (XML) that can be further studied and visualised in a dedicated tool Data Profile Viewer (https://docs.microsoft.com/en-us/sql/integration-services/control-flow/data-profile-viewer). In the task configuration screen (after specifying the destination .xml file location) you can either select Quick Profile to define the data source and set all available profiles to run or you can manually specify which profiles to run in Profile Requests page. Each profile requires a ADO.NET connection (with SqlClient provider fixed), as such the data you need to profile needs to be within an SQL Server instance. There are many possible approaches to the problem of moving data from a flat file to a SQL Server table, for instance you can use

OPENROWSET or BULK INSERT operator in a T-SQL query, you can use SQL Server Import and Export Wizard (an external tool to move data around), flat file import mechanism build in SQL Server Management System (in a database context menu, use Tasks>Import Flat File…), SQL Server Import extension in Azure Data Studio, Tableau Prep (define SQL Server as the output of a flow), SQL Server Integration Services project with proper Data Flow (to move the data around different connections), etc.

Note about Data Profiling task output: "The data profiling components do not include built-in functionality to implement conditional logic in the workflow of the Integration Services package based on the output of the Data Profiling task. However, you can easily add this logic, with a minimal amount of programming, in a Script task. This code would perform an XPath query against the XML output and then save the result in a package variable. Precedence constraints that connect the Script task to subsequent tasks can use an expression to determine the workflow. For example, the Script task detects that the percentage of null values in a column exceeds a certain threshold. When this condition is true, you might want to interrupt the package and resolve the problem before continuing." – for details and examples see https://docs.microsoft.com/en-us/sql/integration-services/control-flow/incorporate-a-data-profiling-task-in-package-workflow?view=sql-server-2017#connecting-the-data-profiling-task-directly-to-an-external-data-source

A Data Quality Assessment (DQA) sheet for source data in the data warehousing process is a critical tool used to evaluate the quality of data before it is integrated into the data warehouse. The sheet outlines the purpose of assessing data quality, which is to ensure that the data entering the data warehouse meets predefined quality standards to support accurate analytics and decision-making. It specifies which data sources are being assessed, the time frame of the data, and the specific data elements or attributes under review. Such a sheet provides some Quantitative Metrics (includes counts of errors, percentage of completeness, accuracy rates, etc.) and Qualitative Metrics (descriptions of data issues, potential impacts, and recommendations for improvement). The final report should provide detailed documentation of what was found during the assessment, including specific issues, their severity, and frequency.

The assessment sheet typically evaluates data against several key dimensions:

- *Accuracy: Checks if the data correctly represents the real-world entities or events it's supposed to model. This might involve comparing data against a known correct source or through validation rules*
- *Completeness: Assesses whether all required data is present. This includes checking for missing values or incomplete records*
- *Consistency: Ensures that data is uniform across different datasets or within the same dataset. This includes checking for conflicting values or formats*
- *Uniqueness: Identifies and quantifies duplicates within the dataset*
- *Timeliness: Evaluates if the data is up-to-date and available when needed, which is crucial for real-time or near-real-time analytics*
- *Validity: Checks if the data conforms to the expected format, type, and range, ensuring it adheres to business rules*
- *Integrity: Verifies the correctness of relationships between data elements, like referential integrity in databases*
- *Reliability: Assesses the consistency of data over time, ensuring it can be trusted for long-term analysis*

In general, a typical approach involves creating two reports – Data Dictionary and QAS. These reports are intended to present your data profiling findings according to the following scope:

- *domain data dictionary – with information about table/sheet/location, attribute name, attribute type (high level type representation, like Numerical, Money, Text, Date, etc.), description (short description of the meaning of the attribute – please provide formatting, enum details). If possible, remember to state formatting restrictions, e.g., post code in "XX-XXX", email in "user@domain", color in "(black, blue, red)".*
- *quality assessment of source data – with information about table/sheet/location, attribute name, attribute type (lower-level type representation, like varchar(20), decimal(5,2), etc.), type of data (nominal, ordinal, interval, ratio, continuous), number of unique values, null ratio, quality assessment description (short description of the results of the attribute quality assessment – focusing on a column consistency assessment, and all issues that might impede using this field in reports/visualisations).*

It is important to remember to mark all occurrences of questionable (in terms of further usage in data analysis and/or processing) attributes. Later, we are going to use a simple data triage system, where red marking is for *attributes that cannot be used further, orange marking is for attributes that require some additional treatment, and green marking is for attributes that can be used directly.*

An exemplar structure of such reports is presented below:

1 – DOMAIN DATA DICTIONARY SHEET

	Location	Attribute name	Attribute type	Description
1	Production.Product	ProductSubcategoryID	Numerical, id	Unique identifier of a product subcategory. References the key in the product subcategory table.
2				

2 – SIMPLIFIED QUALITY ASSESSMENT SHEET

	Location	Attribute name	Attribute type	Type of data	Unique values	Duplicate values	Null values	Quality assessment
1	Production.Product	ProductID	int	nominal	504	0	0	No issues. Coherent and unique values for each product.
2								

***NOTE**: Please remember to provide short description of each issue, note frequency/count of identified issues (or percentage of impacted records) and note the severity level of identified issues (both per issue, and per attribute).*

TASK 4 – DATA PROFILING:

Adventure Cycles, as the leading provider of high-quality bicycles and cycling gear, is actively engaging with its customer base through an online questionnaire. This initiative aims to gather valuable feedback on their products and services to continuously improve the customer experience. The questionnaire is designed to be user-friendly, especially for customers who are already logged into their Adventure Cycles accounts. Upon accessing the questionnaire, users will find some of their basic information, such as their email address and potentially gender, pre-filled, streamlining the process. As users navigate through the questions, they are prompted to provide ratings on various aspects of their interaction with Adventure Cycles, including the website experience, shipping process, and the quality of the purchased product. The questionnaire also seeks an overall satisfaction rating and gathers demographic information to understand their customer base better. The final structure of the collected data for each submitted response is comprehensive, encompassing the following key data points: the date of submission, ratings for the ratingWebsite, ratingShipping, ratingProduct, and ratingOverall, the user's gender, email address, job title, postCode, the source of their interaction (e.g., website direct, advertisement), whether they didPurchase a product, whether they didRecommend Adventure Cycles to others, and users that found the review usefull, and their userAgent string for technical context. This structured approach allows Adventure Cycles to gain insightful data for analysis and decision-making across various aspects of their business. Furthermore, Adventure Cycles also leverages the user's IP address to extract additional location information, which is stored in a separate dataset for further analysis. This allows for a more granular understanding of customer distribution and regional preferences. This structured approach, combined with the supplementary location data, enables Adventure Cycles to gain insightful data for analysis and decision-making across various aspects of their business.

Let us now assume that we've collected the aforementioned information about products' ratings. In short, we have managed to get access to the extracted data (from an external system) – a CSV file (available on EPortal).

Study the provided external file representing an extract from an online questionnaire run by Adventure Works Cycles company focused on rating their services. Present your finding according to the following scope:

- *domain data dictionary – with information about table/sheet/location, attribute name, attribute type (high level type representation, like Numerical, Money, Text, Date, etc.), description (short description of the meaning of the attribute – please provide formatting, enum details). The data itself is rather straightforward, as there are no additional materials regarding the questionnaire please describe your own, individual, interpretation/understanding of the available data. There is no wrong/correct answers for the "Description" field – just make sure it is consistent. If possible, remember to state formatting restrictions, e.g., post code in "XX-XXX", email in "user@domain", color in "(black, blue, red)".*

- *quality assessment of source data – with information about table/sheet/location, attribute name, attribute type (lower-level type representation, like varchar(20), decimal(5,2), etc.), type of data (nominal, ordinal, interval, ratio, continuous), number of unique values, null ratio, quality assessment description (short description of the results of the attribute quality assessment – focusing on a column consistency assessment, and all issues that might impede using this field in reports/visualisations).*

In the resultant tables, please mark all occurrences of questionable (in terms of further usage in data analysis) attributes. Use, red marking for attributes that cannot be used further, orange marking for attributes that require some additional treatment, green marking for attributes that can be used directly, no marking means you are not sure about the attribute (use this last option sparingly). Please remember to later justify your markings and decisions. ***NOTE****: As the solution, please provide details for* *<u>all attributes</u>* *from the provided files. Please remember to present conclusion for each step:*

1 – DOMAIN DATA DICTIONARY

	Location	Attribute name	Attribute type	Description
1	Production.Product	ProductSubcategoryID	Numerical, id	Unique identifier of a product subcategory. References the key in the product subcategory table.
2				

2 – SIMPLIFIED QUALITY ASSESSMENT SHEET

	Location	Attribute name	Attribute type	Type of data	Unique values	Duplicate values	Null values	Quality assessment
1	Production.Product	ProductID	int	nominal	504	0	0	No issues. Coherent and unique values for each product.
2								

NOTE*: Please remember to provide short description of each issue, note frequency/count of identified issues (or percentage of impacted records) and note the severity level of identified issues (both per issue, and per attribute; use basic scale – None, Low, Medium, High, Extreme).*

3 – QUALITY ASSESMENT CONCLUSIONS

Summarize key findings from the assessment, e.g., note Total Records Assessed and Number of Errors Identified, overview Overall Data Quality, and list Key Issues.

Results of this task will be discussed during the laboratory class. Please prepare for a thorough discussion.

RESULTS

After completing the task, double check your results and please create a simple report (use code and comments). Try storing all SQL queries in a single text file (.txt/.sql). Other results should be presented and discussed in a text/Word file. After completing all the tasks, double check your results and please present your results to the teacher. Next, upload files to ePortal.